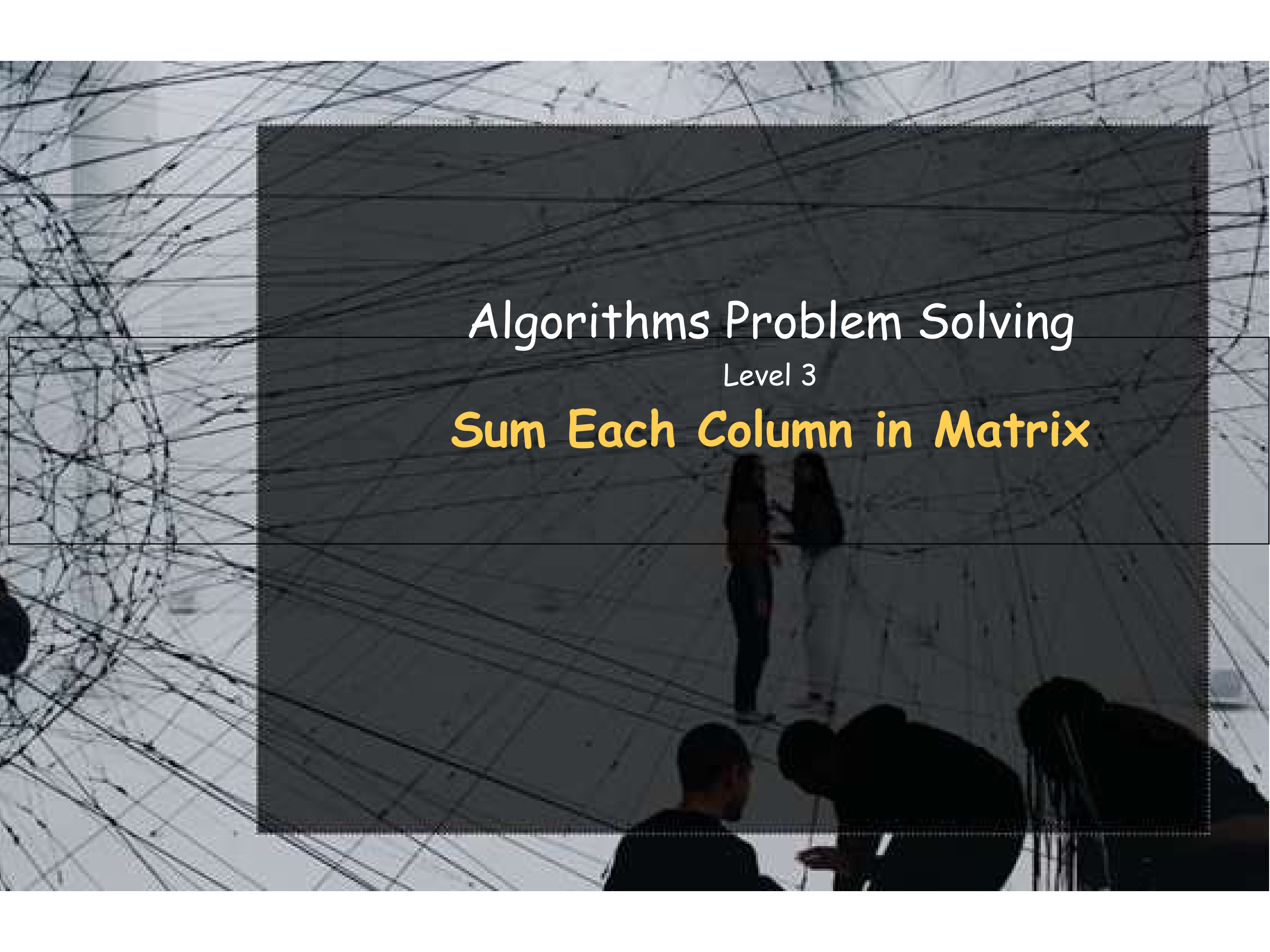


Algorithms Problem Solving Level 3

Sum Each Column in Matrix



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Definition

 Sum Each Column in Matrix

 Program creates a 3×3 matrix of random numbers, then calculates and prints the sum of each column separately.

Explanation

Matrix Structure

 A matrix consists of Rows and Columns.

Accessing Columns

 To access each column, we reverse the loop order: keep the column `j` fixed and move through all rows using `i`.

Calculating Column Sum

 For each column, we add all elements from that column and then print its sum.

Procedures

1. Create a 3×3 matrix using `vector<vector<int>>>`.
2. Fill the matrix with random numbers using `FillMatrixWithRandomNumber`.
3. Print the original matrix using `PrintMatrix`.
4. Calculate and print the sum of each column using `PrintColSums`.

💡 Example

The following is a 3x3 random matrix:

5 46 87

1 12 2

87 2 68

➡ Result

The following are the sum of each Col in the matrix:

Col 1 Sum = 93

Col 2 Sum = 60

Col 3 Sum = 157

Note

◆ For column sums, the column index `j` is fixed while the row index `i` changes.

◆ Example:

 `matrix[0][1]`

 `matrix[1][1]`

 `matrix[2][1]`

◆ These elements all belong to Column 2.

⊘ Warning

⚡ Do not confuse the loop order used for row sums with the loop order used for column sums.



Comparison

Sum Each Row

- ✓ Outer loop uses row `i`.
- ✓ Inner loop uses column `j`.
- ✓ Moves horizontally across each row.
- ✓ Calculates one sum for each row.

Sum Each Column

- ✓ Outer loop uses column `j`.
- ✓ Inner loop uses row `i`.
- ✓ Moves vertically through each column.
- ✓ Calculates one sum for each column.

✦ Summary

- ✓ A 2D matrix contains rows and columns.
- ✓ To calculate row sums, fix `i` and change `j`.
- ✓ To calculate column sums, fix `j` and change `i`.
- ✓ Column elements are added vertically.
- ✓ Nested loops allow us to calculate each column separately.

Thank you 😊